DO NOW — RECALL

Lesson 130 · Symmetric Encryption: One Shared Secret

Name: _____ Class: _____ Date: _____

1. What is the main role of a dictionary token in compression?

2. What is O(n) search complexity associated with?

3. In a BST, which side holds a smaller value?

4. In a queue, what does FIFO stand for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A shorter representation for a repeated substring.

2. Model answer

Linear/sequential search in the worst case.

3. Model answer

The left subtree.

4. Model answer

First in, first out.

By the end, you can…

• Sender has message and shared secret key.

• Algorithm combines plaintext and secret key into ciphertext.

• Recipient uses matching shared key to restore message.

Vocabulary — know these terms

plaintext — Readable original message.

ciphertext — Encrypted output.

symmetric key — Same secret key is used for encryption and decryption.

key distribution — Securely sharing the secret with authorised parties.

NEW LEARNING

Read, annotate and explain

Start plaintext

Sender has message and shared secret key.

Encrypt

Algorithm combines plaintext and secret key into ciphertext.

Decrypt

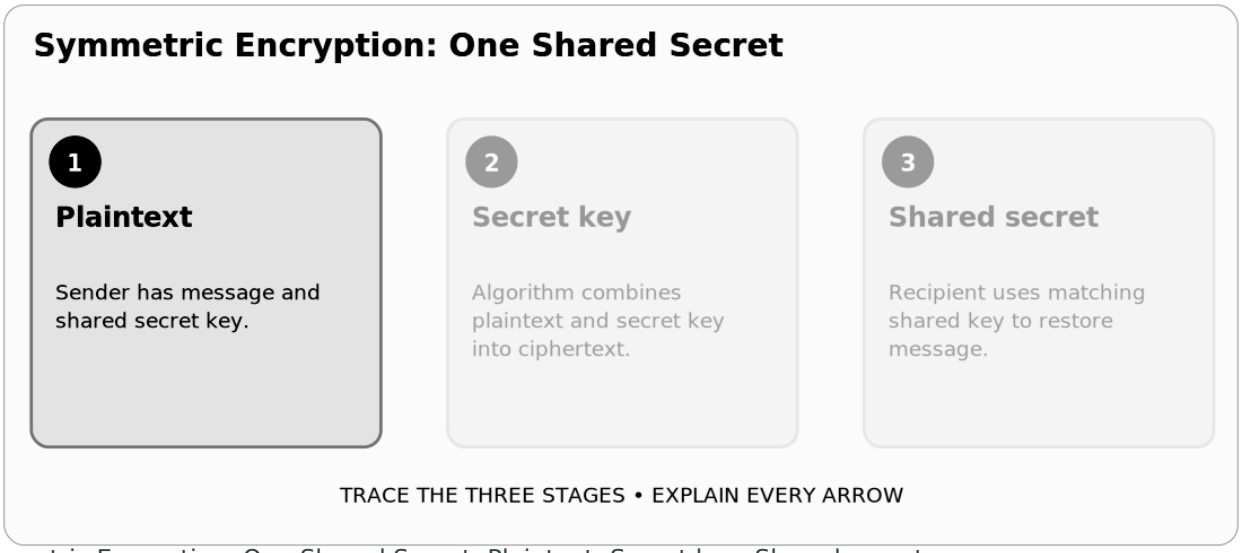
Recipient uses matching shared key to restore message.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

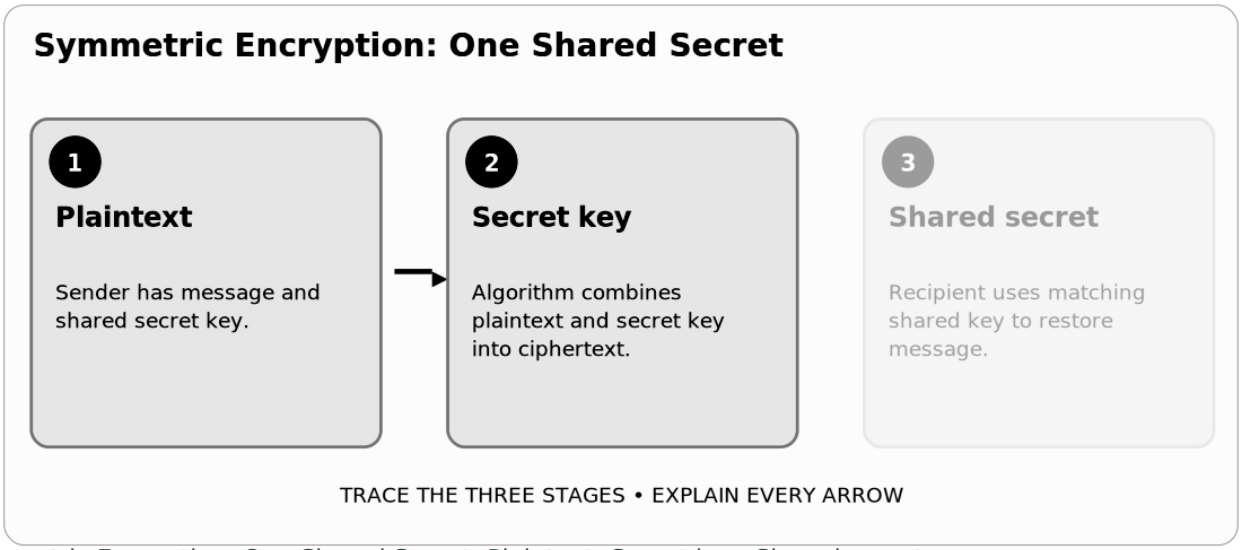
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Symmetric Encryption: One Shared Secret: Plaintext; Secret key; Shared secret

New Learning

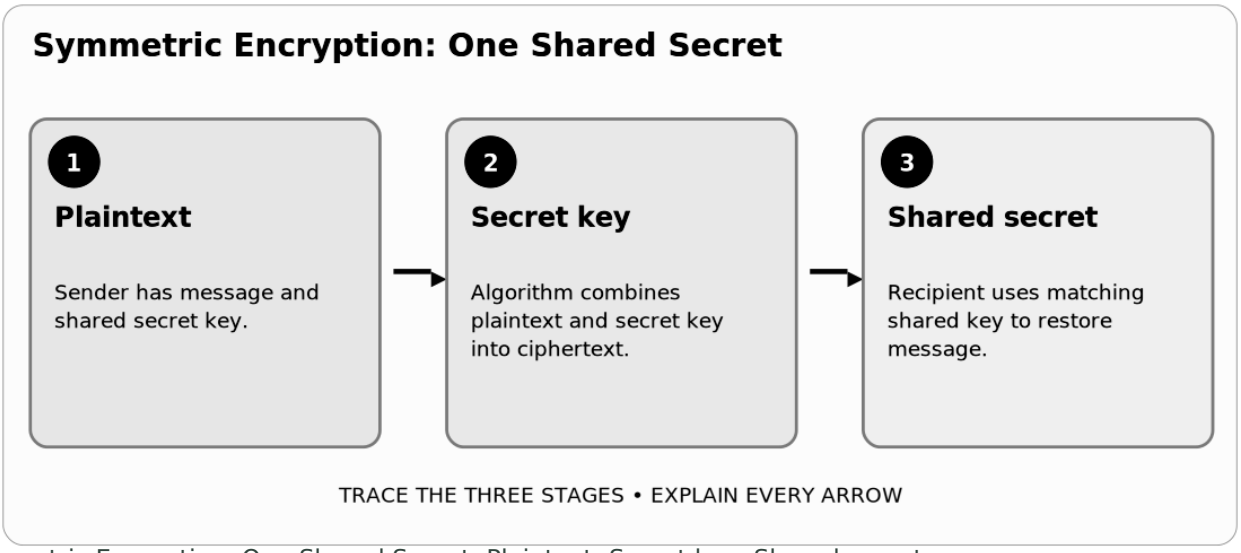


Symmetric Encryption: One Shared Secret: Plaintext; Secret key; Shared secret

New Learning · original lesson diagrams

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New Learning



Symmetric Encryption: One Shared Secret: Plaintext; Secret key; Shared secret

I DO — WORKED EXAMPLE

Follow the worked method

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

Alice: message + key K

Predict the first step from the prompt.

Encrypted ciphertext

Explain the intermediate result or decision.

Bob: decrypt with K

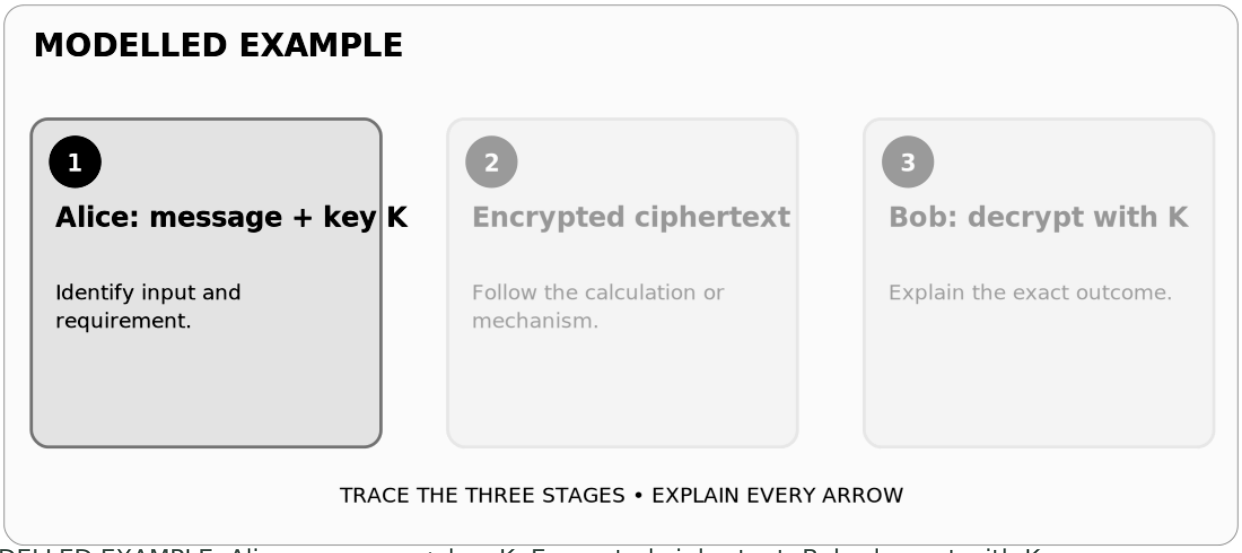
A secure symmetric cipher transforms plaintext using key K. Bob decrypts with the same key; an interceptor without K should not recover plaintext feasibly. Security requires a strong cipher and protected key.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

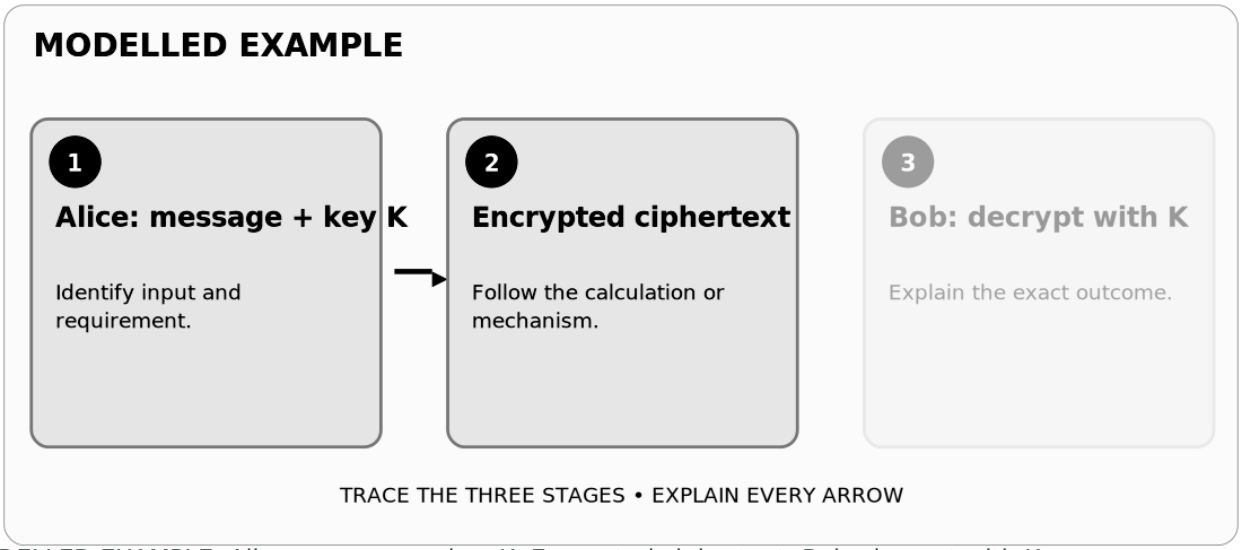
Predict the first step from the prompt.



MODELLED EXAMPLE: Alice: message + key K; Encrypted ciphertext; Bob: decrypt with K

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

Predict the first step from the prompt.



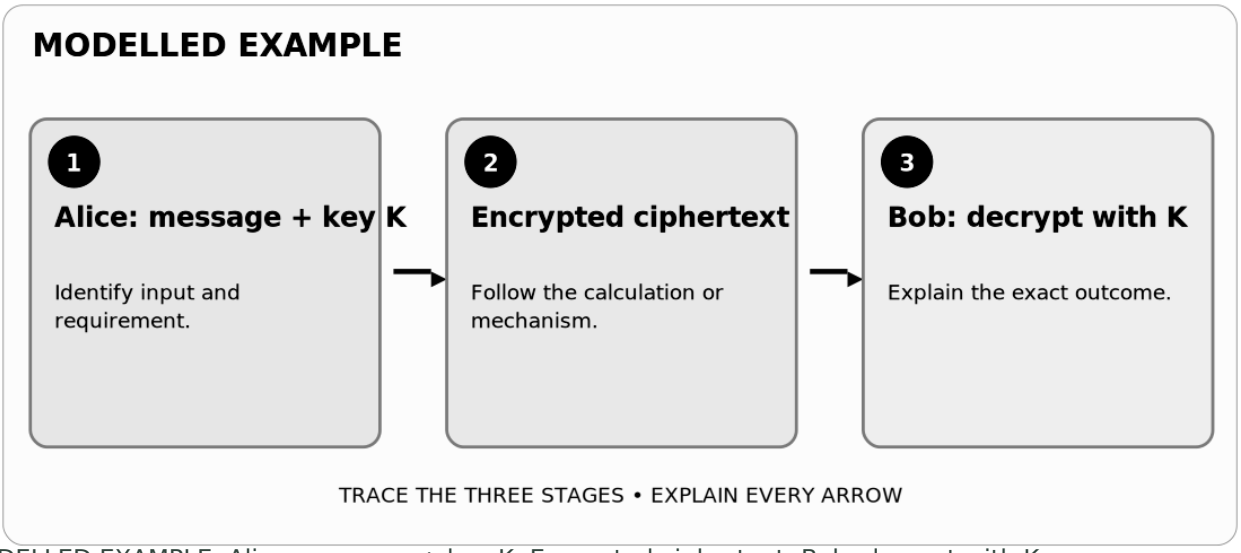
MODELLED EXAMPLE: Alice: message + key K; Encrypted ciphertext; Bob: decrypt with K

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

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MODELLED EXAMPLE: Alice: message + key K; Encrypted ciphertext; Bob: decrypt with K

WE DO — GUIDED PRACTICE

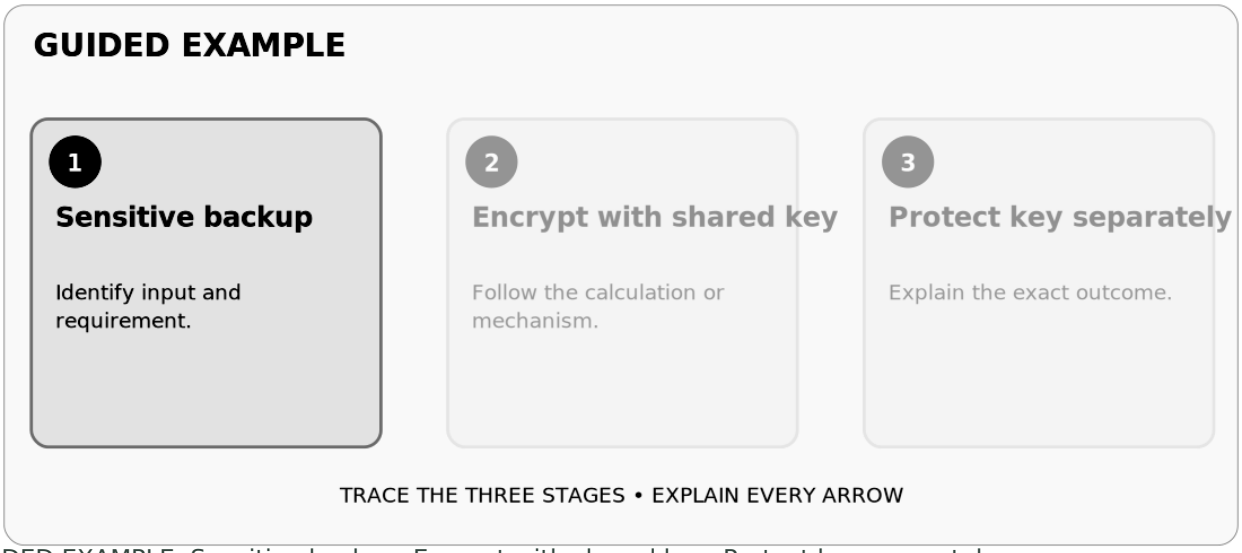
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school backs up a sensitive file to a cloud server. Why might symmetric encryption suit bulk data?

Predict the first step from the prompt.



GUIDED EXAMPLE: Sensitive backup; Encrypt with shared key; Protect key separately

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Sensitive backup Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe the roles of plaintext, key and ciphertext in symmetric encryption. [6]

Question 2

Explain the main key-management challenge in sharing encrypted records between two offices. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate symmetric encryption for a large school backup stored with a third-party provider. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Symmetric encryption uses one public key for encryption and a separate private key for decryption.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

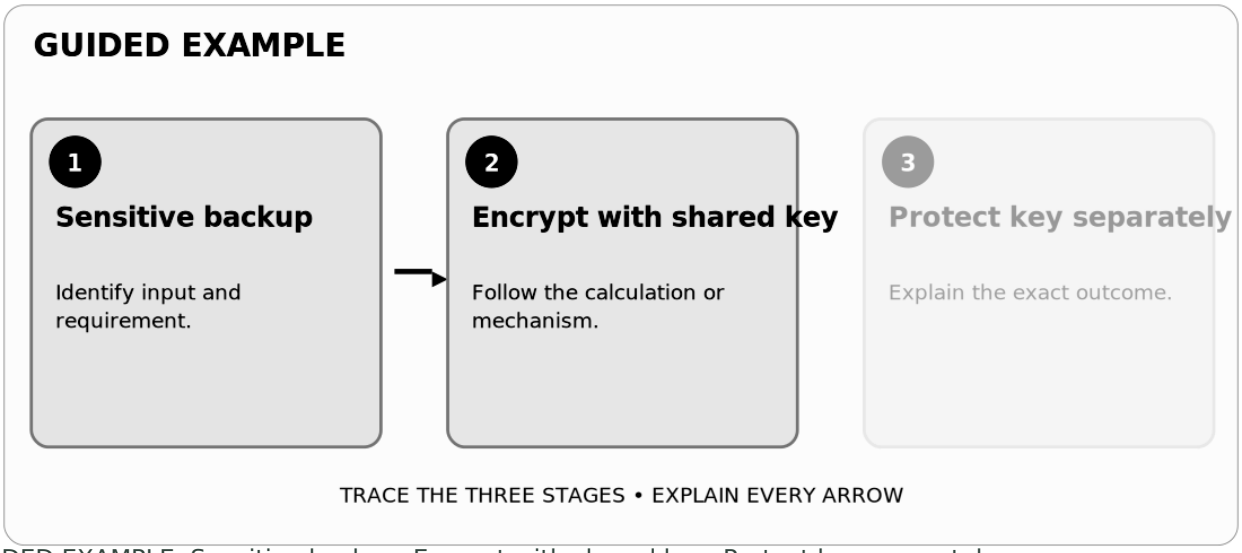
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school backs up a sensitive file to a cloud server. Why might symmetric encryption suit bulk data?

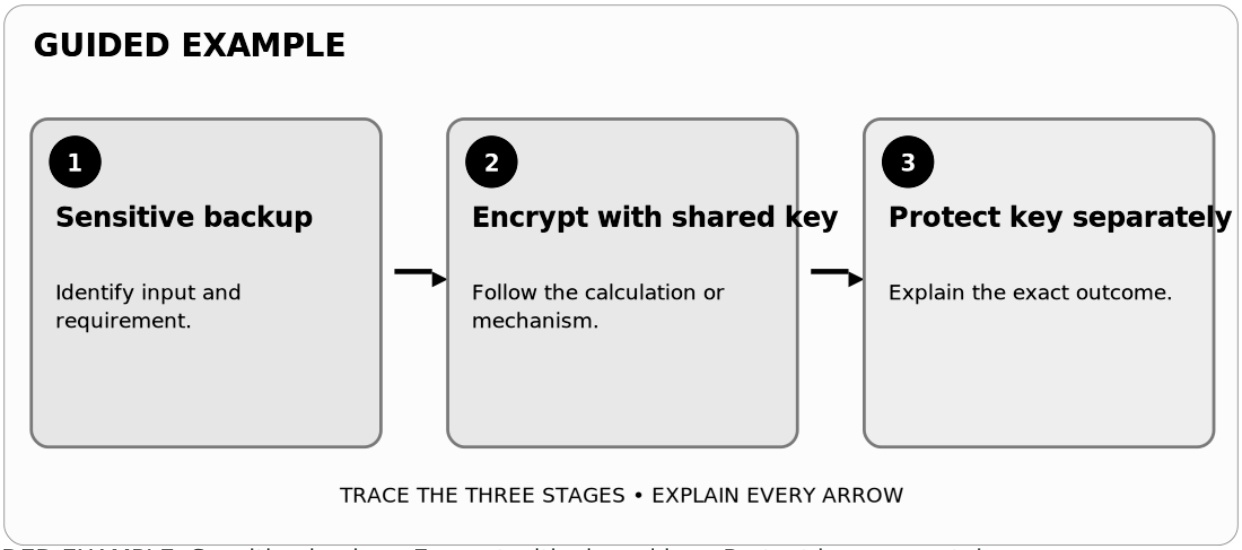
Predict the first step from the prompt.



GUIDED EXAMPLE: Sensitive backup; Encrypt with shared key; Protect key separately

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Predict the first step from the prompt.



GUIDED EXAMPLE: Sensitive backup; Encrypt with shared key; Protect key separately

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Encrypt with shared key

Explain the intermediate result or decision.

Protect key separately

Symmetric algorithms are efficient for encrypting large files, but the encryption key must be stored and shared securely; encrypting a file is not by itself proof of who created it.

IMPROVE: Explain one correction to your own reasoning.